A High-Radix and Parametric Point-to-Point RF Communication System

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Title: A High-Radix and Parametric Point-to-Point RF Communication System

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Cross-Reference to Related Applications

This application is based on the principles disclosed in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Protocol Stack and System for Parametric Data Networking." The present invention applies these concepts to the specific domain of fixed point-to-point radio frequency (RF) links, the contents of which are hereby incorporated by reference.

Abstract

A point-to-point radio frequency (RF) communication system for transmitting data as high-radix symbols that directly represent parametric or symbolic data structures. The system comprises a first transceiver and a second transceiver, forming a fixed wireless link. Each transceiver is equipped with a high-radix RF modulator and demodulator capable of generating and distinguishing N discrete, distinguishable RF states (where N greater than 2) optimized for the specific channel characteristics of the point-to-point link. The system bypasses binary conversion, directly modulating abstract symbols (representing numbers or

operators) onto the RF carrier. This enables the high-fidelity, high-throughput transport of unevaluated parametric data expressions over dedicated wireless backhaul, enterprise, or tactical communication links.

Background of the Invention

Point-to-point RF links, such as microwave backhaul, are a critical component of the global communications infrastructure, prized for their high bandwidth and reliability. However, these links universally operate on a binary basis. Digital baseband processors convert data into a stream of bits, which are then modulated onto an RF carrier using techniques like Quadrature Amplitude Modulation (QAM). While modern QAM schemes can encode multiple bits per symbol, their fundamental purpose remains the transport of a binary data stream. Any high-precision or symbolic data must first be degraded into this binary format, losing its intrinsic fidelity at the source.

Summary of the Invention

The present invention provides a point-to-point RF communication system that natively transmits high-radix and parametric data, fundamentally enhancing both the spectral efficiency and data fidelity of the link. The system abandons the traditional bit-stream-oriented approach in favor of a symbol-oriented one.

The system is composed of a pair of matched transceivers. Each transceiver contains a high-radix RF physical layer. This physical layer is defined by a set of N discrete RF states that are reliably distinguishable over the specific, often line-of-sight, path of the point-to-point link. A symbol-to-state map, optimized for the link's characteristics (e.g., frequency band, distance, atmospheric conditions), directly maps abstract symbols to these RF states.

When transmitting data, the system encapsulates a parametric expression (e.g., a mathematical recipe for a number) into a packet. The transceiver's modulator then directly converts the symbols of this expression into the corresponding sequence of N RF states. The receiving transceiver detects this sequence and demodulates it back into the original, perfect-precision symbolic data. This process eliminates the need for binary conversion and serialization, allowing the full benefits of parametric computing to be extended over high-capacity wireless links.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram of the high-radix point-to-point RF communication system.

FIG. 2 is a conceptual diagram illustrating the mapping of parametric data symbols to optimized RF states for a point-to-point link.

Detailed Description of the Invention

Referring to **FIG. 1**, the point-to-point RF communication system (100) is shown. A first transceiver (110) and a second transceiver (150) form a dedicated RF link (140). The first transceiver (110) receives a parametric data packet (105). Its high-radix RF transceiver unit (120) directly modulates the symbols of the packet onto an RF carrier, generating a high-radix RF signal. This signal is transmitted via an antenna (130). The second transceiver (150) receives the signal via its antenna (160). Its high-radix RF transceiver unit (170) demodulates the signal back into the original parametric data packet (105), which is then passed to the destination system.

FIG. 2 conceptually illustrates the core of the invention. A parametric data structure (210) is composed of abstract symbols. A Link-Optimized Symbol-to-State Map (220) provides a direct correspondence between these symbols and a set of N distinguishable RF states (230). This map is specifically designed to maximize reliability and throughput for the fixed channel characteristics of the point-to-point link. For example, a symbol 'A' is mapped to 'RF State 1', 'B' to 'RF State 2', and so on. The transceiver uses this map to perform the direct symbolic-to-RF conversion.

Claims

What is claimed is:

1. A point-to-point radio frequency (RF) communication system, comprising:
    - a first transceiver, comprising a high-radix RF modulator configured to receive a sequence of abstract symbols and generate a corresponding sequence of N discrete, distinguishable RF states, where N is an integer greater than two; and
    - a second transceiver, communicatively coupled to the first transceiver via a fixed RF link, the second transceiver comprising a high-radix RF demodulator configured to receive said sequence of RF states and decode it back into the sequence of abstract symbols.
2. The system of claim 1, wherein the N discrete, distinguishable RF states are optimized for the specific channel characteristics of the fixed RF link.
3. The system of claim 1, wherein the abstract symbols are components of a parametric data packet, the payload of which is an unevaluated mathematical expression.
4. The system of claim 1, wherein the first transceiver further comprises a symbol-to-state map for converting the abstract symbols to the RF states, and the second transceiver comprises an inverse of said map.
5. A method for point-to-point RF communication, comprising:
    - at a first transceiver, receiving a parametric data packet containing a sequence of abstract symbols;
    - for each abstract symbol, generating one of N discrete, distinguishable RF states based on a predefined mapping, where N is an integer greater than two;
    - transmitting the sequence of RF states over a fixed RF link to a second transceiver; and
    - at the second transceiver, receiving the sequence of RF states and decoding it back into the original sequence of abstract symbols.
6. The method of claim 5, wherein the predefined mapping is optimized for the known channel characteristics of the fixed RF link.
7. The method of claim 5, wherein the abstract symbols represent a complete, unevaluated mathematical expression that is transported across the link without loss of precision.

Drawings

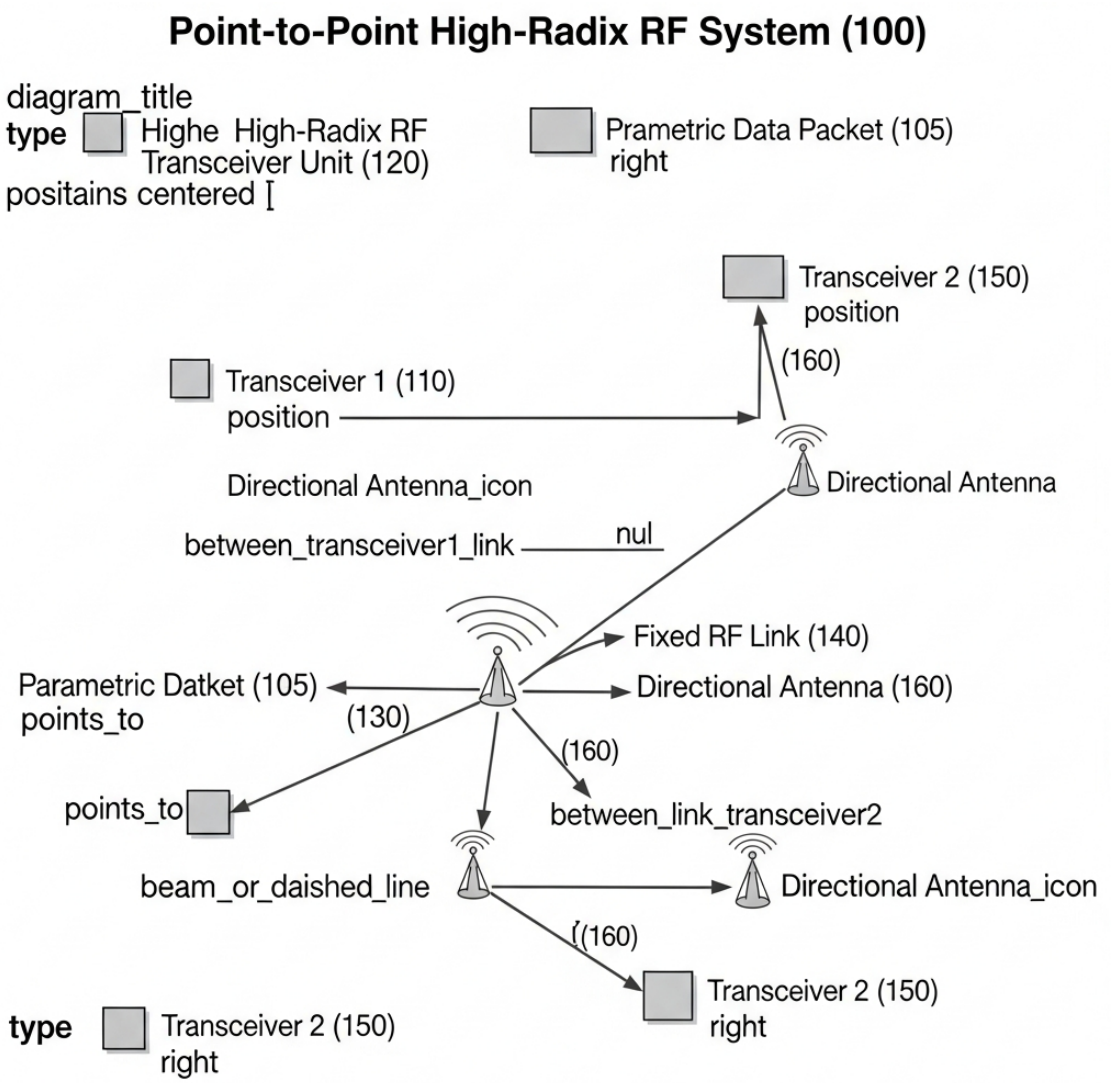


Figure 1: Block diagram of the high-radix point-to-point RF communication system.

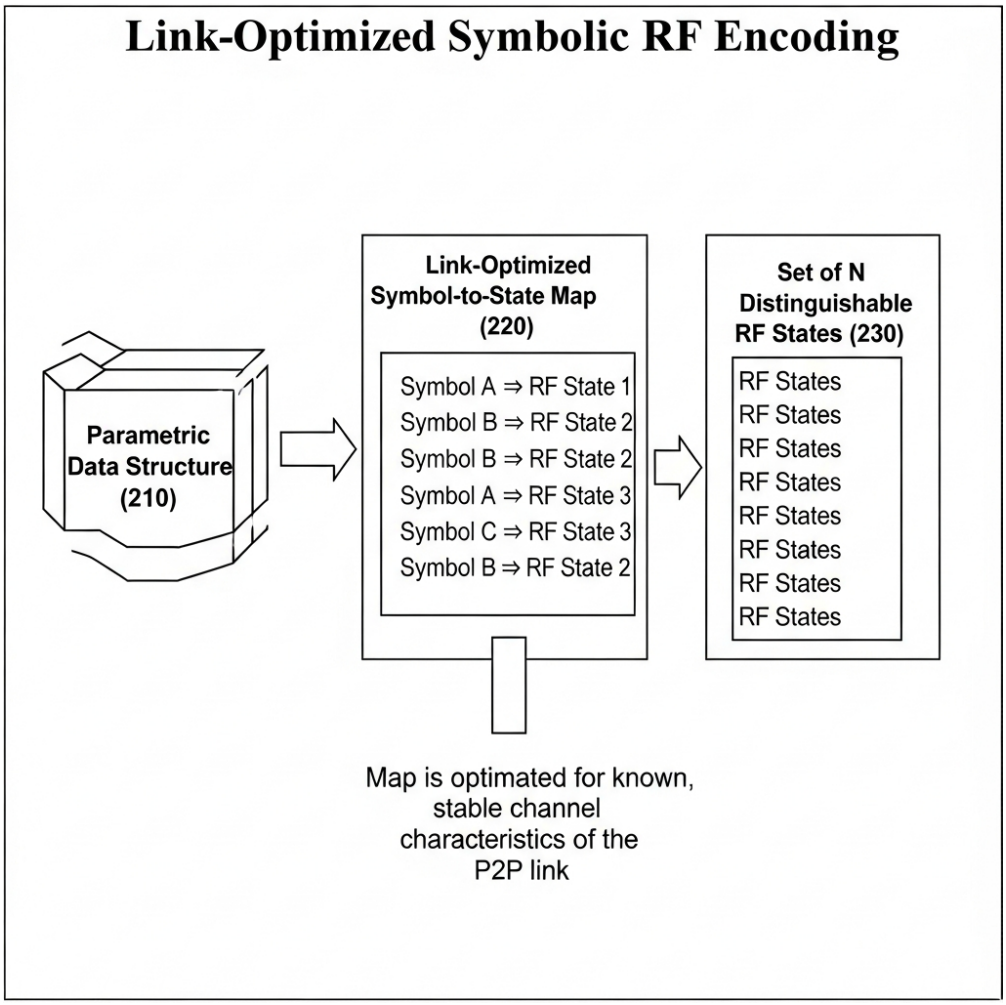


Figure 2: Conceptual diagram illustrating the mapping of parametric data symbols to optimized RF states for a point-to-point link.